Universal Design Review and Recommendations Report

Children's Museum in Oak Lawn

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Universal Design Review and Recommendations Report

Executive Summary

This report presents an evaluation of selected exhibits at the Children's Museum in Oak Lawn using a universal design (UD) framework. Guided by the seven principles of universal design, the evaluation examined how exhibit environments support engagement, usability, and participation for a range of users. A structured observation tool was developed to analyze environmental features, user interaction patterns, and potential barriers within four exhibits: Build It, Gear Wall, Tree House, and What If. The purpose of this evaluation was to identify strengths and areas for improvement to support broader participation in play-based experiences.

Findings indicated several strengths across exhibits, particularly in the use of open-ended materials that supported flexible engagement, creativity, and collaborative play between children and caregivers. Many exhibit components allowed for multiple users to interact simultaneously, encouraging social participation. However, patterns across exhibits also highlighted opportunities to strengthen overall access to play. These included reliance on caregiver assistance, limited cues to support independent play, and variability in physical demands required to engage with exhibits.

Based on these findings, recommendations were developed to support independent participation, enhance clarity of engagement, and promote more consistent access across exhibit environments. Key strategies include strengthening communication, adjusting physical display of components, and maintaining flexible, open-ended play opportunities while improving entry points into engagement. Longer-term considerations include expanding communication supports through multi-modal resources and incorporating universal design principles into future exhibit planning. Collectively, these recommendations aim to build upon existing strengths within the

museum while supporting continued development of environments that foster meaningful, engaging play experiences for a broad range of visitors.

1. Introduction

The Children's Museum in Oak Lawn (CMOL) provides play-based learning experiences designed to support engagement, exploration, and development for children and families. As play is widely recognized as a fundamental occupation of childhood and a key contributor to participation, it is important that play environments are designed to support a range of users. Universal design offers a framework for creating environments that are usable by all individuals to the greatest extent possible, without requiring adaptation.

Within children's museum settings, applying universal design principles may support increased participation, independent engagement, and shared play experiences among children with diverse abilities. However, variability in exhibit design may influence how children interact with and access play opportunities. This report presents an evaluation of selected exhibits using a universal design lens to better understand how environmental features support or limit engagement. Findings from this evaluation are used to inform recommendations aimed at strengthening access, usability, and participation across exhibit environments.

2. Purpose of Review

Children's museums are designed to support exploration, creativity, and learning through play. However, environments intended for broad public participation do not automatically provide equitable access for all children and families. Research suggests that children with disabilities may encounter physical, social, and communicative barriers that restrict their ability to fully participate in play settings, with play environments that often lack accessibility, usability, and support for peer interaction (Prellwitz & Skär, 2007). Within occupational therapy (OT),

occupation is defined as the everyday activities that individuals engage in that are meaningful and purposeful (Baum et al., 2015). OT offers a distinct perspective on play by conceptualizing it as an occupation that supports children's development and participation in everyday life. As play is widely recognized as a fundamental occupation of childhood and an important contributor to development and participation, ensuring that play environments support children with a range of abilities is an important consideration for community spaces (Smet & Lucas, 2015; James et al., 2022).

The Children's Museum in Oak Lawn (CMOL), which serves a diverse community of children and families, has identified accessibility, inclusivity, and family engagement as important priorities for its programming and exhibit development (A. Woodworth & H. Hendrickson, personal communications, January 22, 2025). Although CMOL does not have a formalized definition of play, its philosophy is consistent with Gray's (2017) conceptualization of play as self-chosen, self-directed, intrinsically motivated, and imaginative, with the acknowledgement that not all museum-based activities embody these characteristics (H. Hendrickson, personal communication, March 28, 2025).

Universal design (UD) provides a useful framework for examining how environments can support broader participation. Rather than relying solely on accommodations after barriers arise, UD promotes the design of environments that can be used by the widest range of individuals possible without the need for adaptation (Connell et al., 1997). In the context of UD, OT involvement in the built environment can be conceptualized as maximizing *occupational performance*, which is the result of dynamic interactions between a person, their environment, and their occupation (Young et al., 2019; Baum et al., 2015). In museum contexts, this perspective extends beyond reaching minimum accessibility requirements by encouraging

consideration of multiple dimensions of engagement, including physical access, sensory information, cognitive clarity, and social participation. This approach extends beyond addressing accessibility standards to support greater opportunities for engagement within the environment.

Although UD principles have been explored across a variety of built environments, including playgrounds and public spaces, there is comparatively limited literature examining how these principles are applied within children's museum exhibits specifically. As a result, museums seeking to improve participation for all children may have limited practical tools for evaluating existing exhibits or identifying opportunities for improvement. UD is further explored in the next section.

The purpose of this review was to examine the results of an evaluation conducted on exhibits selected through a needs assessment, which included stakeholder surveys and observations, at the CMOL through a UD lens in order to identify strengths, barriers, and opportunities for improvement related to play and participation. By systematically reviewing exhibit environments using a structured observation tool informed by universal design principles, this review aimed to provide practical insights to support ongoing efforts to promote increasing play opportunities for all children.

Grounded in the Person Environment Occupation Performance (PEOP) theory, this review and the processes of evaluation reflect the understanding that effective participation in occupations arises from the dynamic interaction between the person, environment, occupation, and performance (Baum et al., 2015). In the context of this project, the occupation of interest is play; the museum environment serves as an important context that can either support or limit children's engagement in play activities. Examining exhibit environments through this lens

allows for consideration of how environmental features influence participation, independence, shared engagement with caregivers, and opportunities for meaningful play.

3. Overview of Universal Design

Universal design refers to the design of products, environments, and experiences so that they can be used by the widest possible range of individuals, regardless of age, ability, or background, without the need for specialized adaptation (Connell et al., 1997). UD originated within architecture and was originally coined by Ron Mace as an approach to creating houses that are accessible and can promote aging in place; it has since expanded into broader contexts and environments, including education and community settings, as a framework for supporting participation across diverse users (Centre for Excellence in Universal Design, n.d.-a, n.d.-b). Over time, UD has been increasingly applied across educational, recreational, and community settings as a framework for promoting increased participation.

UD encourages designers and organizations to anticipate variability in users' abilities, preferences, and experiences when designing environments. In practice, this means considering factors such as physical access, types of perceptible information, flexibility of interaction, and tolerance for error during the design process. Universal design has been associated with improved participation by enhancing the fit between individuals and their environments (Young et al., 2019).

The framework is commonly described through seven guiding principles originally written by a team led by Ron Mace in the North Carolina State University that includes: equitable use, flexibility in use, simple and intuitive use, perceptible information, tolerance for error, low physical effort, and size and space for approach and use (Centre for Excellence in Universal Design, n.d. -b; Connell et al., 1997). Together, these principles provide a structure for

examining how environments support usability across a range of user needs. A description of each of these principles and how they were interpreted in the context of play during evaluations is outlined later in the findings section.

Within children's museums and other informal learning environments, UD has particular relevance because exhibits are intended to support open-ended exploration and engagement for diverse audiences. Design guidance developed for museum and science center environments emphasizes that exhibits should provide multiple ways for visitors to interact with content, communicate information through visual, tactile, and auditory channels, and allow individuals to engage at varying levels of complexity (NISE Network, 2010). These approaches align with broader design frameworks that promote environments capable of supporting both independent exploration and shared participation among all children and caregivers.

Several practical tools and resources have been developed to support organizations in evaluating environments through a universal design perspective. Examples include the Smithsonian Accessibility Guidebook and the state of New York's universal design guidelines, which provide structured criteria for examining accessibility features within built environments (Smithsonian Institution, n.d.; Levine, 2003). These tools highlighted the assessment of environmental elements such as reach ranges, physical circulation space, perceptible cues, and clarity of instructions that may influence participation for individuals with diverse abilities. However, these resources lacked the specific application to indoor play environments and the application of these principles specifically to play.

For the purposes of this project, the seven universal design principles served as a conceptual framework for examining exhibit environments within CMOL. These principles

informed the development of the observation tool used during the review process and guided the organization of findings presented later in this report.

4. Methods

4.1 Universal Design Review Tool

Observations were guided by a structured universal design review tool developed for this project. The UD Tool was developed through an iterative process informed by multiple design guidance documents specific to museum, exhibition, and built environments, guided by the seven principles of UD. Foundational structure for the tool was drawn from the UD principles (Connell et al., 1997), which were operationalized into observable exhibit-level criteria using guidance from Universal Design for Interactives in Museums (Reich, 2005), the NISE Universal Design Guidelines (NISE, 2010), the Universal Design New York (UDNY) report and embedded checklist (Levine, 2003), and the Smithsonian accessibility guidance (Smithsonian Institution, n.d.). Each source contributed distinct elements to the tool's development. Reich (2005) and the NISE guidelines (2010) informed criteria related to user interaction, multimodal engagement, and variability in participation, particularly within informal learning contexts. The UDNY checklist (Levine, 2003) supported the translation of broader UD principles into concrete, observable environmental features, including considerations related to spatial layout, reach, and physical access. The Smithsonian guidance (Smithsonian Institution, n.d.) further contributed to criteria addressing sensory accessibility, communication supports, and exhibit usability across diverse visitor needs. Additional elements of the tool were informed by the author's synthesis of the broader universal design literature related to environments and play, reflecting concepts that emerged across sources but cannot be attributed to any single document.

Following initial development, the tool was refined through multiple rounds of revision to improve clarity, usability, and alignment with UD principles. Early revisions focused on simplifying item wording, reducing ambiguity, and ensuring that all items were framed such that the presence of a feature reflected greater alignment with universal design. Additional revisions were made following expert feedback received from the Children's Museum of Indianapolis, including an occupational therapist, the Accessibility Manager, and the Director of Research and Evaluation, which highlighted the need for clearer operational definitions, improved consistency across principles, and greater specificity in areas such as perceptible information and physical effort. These refinements resulted in a tool that balances conceptual alignment with UD and practical application within a children's museum's exhibit evaluation context.

The review tool is in a checklist format and includes a series of observational prompts aligned with each universal design principle. It also allows the reviewer to optionally indicate whether specific environmental features are present, partially present, not present, or not applicable within a given exhibit. The tool was designed to support descriptive observation rather than quantitative scoring. Subsequent to its initial application, the tool was refined through incorporation of additional feedback and insights gained from its use within the museum environment.

4.2 Exhibits Reviewed and Observation Procedures

This review focused on four exhibits located within the museum: Build It, Gear Wall, Tree House, and What If. These exhibits were selected based on a survey conducted within the museum environment. Build It and Gear Wall were popular responses under the categories of exhibits that could use improvement and/or exhibits that were underutilized. What If was selected from structured interviews with staff. Tree House was selected due to its popularity with

children. Each exhibit includes opportunities for hands-on interaction and exploratory play, making them appropriate contexts for examining how environmental design influences children's participation and engagement.

Observations were conducted within the natural museum environment during regular hours of operation. Observations were conducted through short, naturalistic observation sessions while the exhibits were in use by museum visitors. Each exhibit was observed for two sessions each of which lasted approximately 10-15 minutes and focused on how children and caregivers interacted with the exhibit environment.

During these sessions, attention was directed toward environmental features that could influence participation, such as reach ranges, physical access to exhibit components, clarity of instructions, sensory information, and opportunities for varied interaction styles. Observations focused on the environment and patterns of interaction rather than on evaluating individual children or families. Multiple observation sessions were conducted for each exhibit in order to capture a range of interactions and environmental conditions. Observations of each exhibit's play time were paired with overall assessment of the exhibit space for items that did not require engagement to be assessed (e.g. "A clear turning space (approximately 60 inches in diameter or equivalent) is available to allow individuals using mobility aids to turn or reposition independently.")

4.4 Data Documentation

Observations were documented using the tool alongside written notes describing environmental features and patterns of interaction observed during each session. These notes were used to capture contextual information that may not have been fully reflected within the checklist items. These notes are optional and are not required to complete the tool itself.

The collected observation data were later reviewed and organized according to the seven UD principles to support identification of patterns across exhibits. Findings from this process are presented in the following sections of the report.

5. Key Findings

Observations conducted across the selected exhibits revealed several patterns related to how environmental design features may support or limit participation in play activities. Findings are organized according to the seven principles of universal design in order to highlight how exhibit environments align with or diverge from these principles. While individual exhibit observations informed these findings, patterns were most meaningfully understood through cross-exhibit analysis, with specific examples included to illustrate how these patterns manifested within particular exhibit contexts. Appendix B contains tables that organize findings into ratings per principle and a summary table. Individual exhibit results can be found in the accompanying material which includes the four completed evaluation documents.

5.1 Equitable Use

Equitable use relates to the extent to which an exhibit can be accessed and used by a wide range of individuals without requiring adaptation. Within a play environment, this could be reflected in whether children with diverse abilities, including differences in age, height, cognition, and gender, can engage with the same materials and activities in similar or parallel ways. The main question is whether all children can engage with the intended core activity.

Across the exhibits observed, many play opportunities supported open-ended engagement and collaborative participation between children and caregivers. Several exhibit components allowed multiple users to interact with materials simultaneously, encouraging shared play and social interaction. However, certain elements required specific physical abilities, such as

reaching elevated components or accessing interactive elements positioned outside of typical child reach ranges. This was particularly evident within the Gear Wall, Tree House, and portions of the Build It exhibit, where children often relied on caregiver assistance to fully engage with the activity. Increased reliance on caregivers was also observed in the Gear Wall, What If, and Build It exhibits, which may limit opportunities for independent exploration and sustained engagement, particularly when caregiver support is divided or unavailable. In exhibits where fostering adult participation with children is an intended outcome, this reliance may align with design goals. However, in the context of supporting broader opportunities for independent initiation and sustained engagement, this pattern remains an important consideration. Physical access to exhibit components also varied across locations. Exhibits such as Build It and Tree House generally provided more accessible open layouts that supported approach and interaction, while What If and Gear Wall exhibits presented more constraints related to reach, positioning, or spatial access. These variations suggest that opportunities for equitable participation were not experienced consistently across exhibits.

5.2 Flexibility in Use

Flexibility in use refers to the ability of a space or activity to accommodate a variety of preferences, abilities, and approaches. In the context of play, this may be observed in exhibits that allow children to interact in multiple ways, adjust how they engage, or use materials according to their own ideas.

Across the observed exhibits, materials often supported varied engagement, including exploratory manipulation, structured problem-solving, individual play, and collaborative interaction with peers or caregivers. In the Tree House exhibit, flexibility in play was highly present due to its open layout with varied options for engagement. Despite these strengths,

certain interaction pathways appeared more dominant in specific contexts. In the Gear Wall exhibit, the vertical layout and component placement appeared to guide children toward a more limited range of interaction styles, primarily focused on arranging and activating gear systems. While this did not prevent engagement, it may have reduced opportunities for alternative approaches to play. Providing additional cues that are widely accessible may further enhance flexibility by encouraging varied interaction styles and supporting a broader range of users.

5.3 Simple and Intuitive Use

Simple and intuitive use relates to how easily a user can understand how to engage with an exhibit, regardless of prior experience or ability. Within play environments, intuitive use refers to the extent to which users can identify how to begin and sustain interaction without reliance on explicit instruction. While exhibits are often designed to promote growth and learning, having highly intuitive and simple play environments may not always be feasible. As such, identifying areas where intuitive use is limited is important for informing strategies that enhance overall usability

Across exhibits, children frequently relied on exploration rather than explicit instructions to initiate play. While exploratory learning aligns with the goals of children's museums, younger age groups appeared to experience greater difficulty in more open-ended exhibits when accessible guidance points were not present. In exhibits such as Build It and Gear Wall, children often required caregiver support to initiate or continue engagement, suggesting that entry points into play were not always immediately clear. In contrast, intuitive use was more consistently observed in the Tree House exhibit, where children readily identified opportunities for interaction within the structured environment and engaged with climbing elements and musical features without hesitation. In the What If exhibit, although written guidance was present and

appeared to target certain reading levels, the environment itself did not consistently invite engagement across users. Across exhibits, the absence of clear or alternative entry points contributed to hesitation or reliance on caregiver assistance. Incorporating visual or physical cues may further support independent initiation and sustained engagement.

5.4 Perceptible Information

Perceptible information involves how effectively an exhibit communicates necessary information through different sensory channels to support use and engagement by a wide range of users. Across the observed exhibits, visual and tactile elements such as color, texture, and movement played a strong role in attracting attention and encouraging interaction. However, the ways in which information about how to engage with exhibit components was communicated varied between exhibits. In some cases, particularly within the What If exhibit, information was communicated primarily through text-based instructions, which may not be usable for all visitors. Each of the main parts of What If (Ways to Hear, Ways to See) was paired with an activity that was flexible in use across age groups; however, these activities required adult support to initiate interaction. The layout of the space further influenced engagement with the enclosed poster displays and with limited non-text materials available to attract users. In other exhibits, children relied on observation, trial-and-error, or caregiver explanation to determine how to interact with materials. Expanding the use of multimodal communication strategies, including visual cues, tactile indicators, and other sensory supports, may strengthen understanding and promote engagement across a wider range of users.

5.5 Tolerance for Error

Tolerance for error relates to how well a design minimizes negative consequences of mistakes or unintended actions. Across the exhibits observed, opportunities for trial-and-error

learning were consistently present. Children were able to experiment with materials without risk of damaging the exhibit or causing harm. Components were generally resettable, reusable, and adaptable, which supported iterative problem-solving and continued engagement. A primary point to note in this stage is that there were limited to no activities that caused users to “fail” or stop engagement. This enables children to play in exhibits at their own pace and can promote child-led learning. These approaches appeared to encourage creativity and persistence, as children frequently repeated interactions in order to explore different outcomes.

5.6 Low Physical Effort

Low physical effort refers to the degree to which an activity can be performed comfortably and without excessive exertion. Within play environments, it is important to recognize that some activities are intentionally physically demanding to support growth and development. As such, for the purposes of this evaluation, low physical effort was assessed in relation to whether opportunities were available for children to engage with exhibit components without requiring excessive physical effort beyond what was intended, and whether opportunities for rest or modulation of effort were incorporated into the activity.

The physical effort required to interact with exhibit components varied across activities. Some elements were easily manipulated and supported sustained engagement, while others required greater strength, lifting, or reaching. Build It was an exhibit where high modulation of effort was seen, with children observed to be pacing independently, and choosing their level of physical engagement. Tree House was an exhibit that was required higher physical effort, particularly for younger age groups; however, there were observed instances of children taking seated rest breaks on stairs or returning to caregivers for support. In instances exhibits required more physical effort than children could produce, they often relied on caregivers for assistance

with repositioning materials or accessing components, particularly evident in the Gear Wall. The Gear Wall received a lower rating for low physical effort due to the sustained reaching, lifting, and force required to manipulate components, particularly given its vertical layout. While effortful play is an intended and meaningful aspect of engagement, limited alternative access points or reduced-demand interaction options may restrict participation for children with varying physical abilities. Considering adjustments or providing alternative interaction options may help broaden participation by supporting children with a wider range of physical abilities.

5.7 Size and Space for Approach and Use

This principle considers whether the physical space allows for access, movement, and interaction by individuals with a range of body sizes and mobility needs. Across exhibits, some areas, like Build It and Tree House, provided open floor space that allowed children and caregivers to approach materials from multiple directions, supporting collaborative play and flexible positioning. However, during periods of higher visitor activity, certain areas, like the Gear Wall and What If, became congested, which occasionally limited users' ability to comfortably approach or interact with exhibit components. The What If exhibit is positioned within a relatively constrained area, where certain corners and sections appear clustered and limit use for individuals who utilize mobility aids. Consideration of spatial layout, position of exhibits, and circulation patterns may support more consistent access and use across varying usage conditions.

6. Key Themes

A thematic analysis was conducted to identify patterns across exhibits in relation to universal design principles and participation outcomes. Themes were developed through iterative review of observational data, item-level patterns identified through the evaluation tool, and comparisons across exhibits. These themes reflect recurring conditions that shaped participation

and provide insight into both strengths and areas for further consideration within the museum environment. A visual matrix representing the themes and corresponding relationships are presented in Appendix A.

The five themes presented in this section captured the most consistent and meaningful patterns across the evaluated exhibits. They represented conditions that shaped participation in ways that were relevant to the museum's commitment to creating an accessible and inclusive space for everyone. Together, these themes help explain where design strengths are already present, where barriers are more likely to emerge, and how those patterns may inform future efforts to strengthen UD within exhibit environments.

6.1 Inconsistent Access Across Exhibits

A primary pattern identified across the review was that access was not experienced consistently from one exhibit to another. Although all exhibits were located within the same museum context, the degree to which they supported approach, interaction, and engagement varied. In some spaces, such as Build It and Tree House, materials were arranged in ways that supported flexible entry and collaborative interaction. In others, including the Gear Wall and What If, usability depended more heavily on factors such as height, reach, physical positioning, cognitive ability, or the ability to interpret how the exhibit functioned. This theme was most closely associated with principles related to equitable use, low physical effort, and size and space for approach and use. The significance of this finding lies in the unevenness of access rather than the presence of isolated barriers. When participation conditions vary across exhibits, engagement becomes more dependent on environmental design than on a child's interests or preferences.

In responding to this theme, it is not appropriate to frame inconsistent access as an issue that can be resolved through a single solution. Rather, it reflects an ongoing consideration for the

museum as it continues to evolve. Supporting more consistent access and reach across a range of users should be integrated into future decision-making and exhibit development. As the museum grows as a community organization, maintaining consistency in how play experiences are supported across exhibits may help reduce variability in participation. To address current inconsistencies, targeted recommendations are provided in the following section.

6.2 Barriers to Independent Engagement

A second theme related to the extent to which exhibit design supported or constrained independent initiation of play. Across exhibits, children did not consistently appear able to approach materials and begin interacting without assistance. In some cases, this was related to physical demands, such as reaching or manipulating components, as observed in the Gear Wall and Tree House. In other cases, barriers were related to unclear entry points or lack of intuitive cues, as seen in Build It and What If. These conditions did not necessarily prevent engagement altogether, but they often shifted the responsibility for initiating play from the child to the caregiver.

Caregiver involvement is not inherently problematic and is often central to museum play. However, when adult intervention is required because of design barriers rather than because of social choice or shared exploration, the opportunity for self-directed play may be reduced. For the museum, this theme suggests that supporting participation is not only a matter of allowing access to a space, but also of making entry into play sufficiently clear and manageable that children can begin engaging on their own terms. This theme highlights the importance of designing environments that support independent initiation of play while still allowing for shared experiences.

6.3 Limited Multimodal Communication Supports

Another pattern that emerged was the limited use of multimodal communication supports across exhibits. In several cases, children relied on observation, trial-and-error, or caregiver explanation to determine how to engage with materials. This was particularly evident in the Gear Wall and Build It exhibits. While exploratory learning is an important component of play, the absence of clear visual, symbolic, or tactile cues may limit opportunities to engage for children who benefit from additional forms of environmental guidance. This theme emphasizes the role of communication as a design feature and highlights the importance of providing information through multiple sensory and cognitive channels.

6.4 Physical Design Influencing Participation

The physical configuration of exhibits consistently influenced how children were able to participate. Environmental demands related to height, force, positioning, and movement shaped engagement, particularly within the Gear Wall and Tree House exhibits. Activities that required reaching, climbing, or sustained effort created conditions that may privilege some users while limiting others. This pattern aligns most closely with the principles of low physical effort and size and space for approach and use and is distinct from inconsistent access across exhibits, as it reflects the demands of the interaction itself rather than variation between environments. Considering these factors is important for understanding how children with varying body sizes, endurance levels, strength, and mobility patterns may experience the same environment differently. This has implications for how participation is supported, as differences in physical demand may influence not only who is able to engage, but also how long, how independently, and in what ways engagement is sustained.

6.5 Flexible Design Supporting Engagement

A key strength identified across the review was the role of flexible, open-ended design in supporting engagement. Exhibits such as Build It and What If allowed children to interact with materials in multiple ways, supporting a range of play styles and levels of complexity. In these environments, children were able to enter play through different pathways and engage according to their own interests and abilities. This flexibility appeared to support sustained engagement and a broader range of participation behaviors. This theme highlights the value of adaptable, open-ended environments as a key feature of inclusive play design.

This theme was most strongly represented in the Build It and What If exhibits and aligned most closely with the principles of flexibility in use and tolerance for error. Its importance lies in the fact that it highlights what appears to be working well. Flexible design did not simply reduce barriers; it also expanded possibilities for participation. For the museum, this theme offers a useful direction for future exhibit planning because it identifies a design quality that appears to support inclusion without requiring rigid standardization. Open-ended, adaptable environments may be particularly well suited to museum contexts because they allow children to engage according to their own abilities, interests, and pace.

8. Recommendations

The following recommendations were developed based on patterns identified across the evaluated exhibits. While observations were conducted within four specific exhibit spaces, several recommendations may be applicable more broadly within the museum environment. Recommendations are presented from those requiring relatively minimal modification to those involving more substantial environmental considerations. Collectively, these strategies aim to strengthen access for all users, support independent participation, and enhance play opportunities through the application of universal design principles.

One key opportunity identified across multiple exhibits involves strengthening the use of visual and multimodal communication to support understanding of how to engage with exhibit components. Observations indicated that children frequently relied on caregivers or peer modeling to determine how materials could be used, suggesting that entry points into play were not always clearly communicated through the environment itself. Incorporating universal symbols, icons, or visual representations of possible interactions may support clearer initiation of play without relying on written instructions. These supports can help communicate relationships between components while maintaining the open-ended nature of the activity.

While there may be concerns that adding visual supports could make play more rigid or prescriptive, these strategies do not define a single way of interacting with the exhibit. Rather, they provide an accessible starting point that can be adapted, ignored, or expanded upon by the user. In this way, visual supports can enhance engagement without limiting creativity, supporting both independent initiation and flexible exploration. Minimizing reliance on text through the use of universal symbols and images may better support a broader range of users. When text must be used, language that invites exploration should be prioritized, to prevent rigid play. Terms such as "try this," "here's an idea," or "what can you do" are more open and flexible than phrases such as "follow these steps," "do this first," "correct way," "you should," or "complete the activity by…" In some cases, providing a single example of use may be beneficial. For instance, within the Gear Wall exhibit, a pre-assembled set of gears could demonstrate how components interact, offering an initial point of engagement while still allowing for exploration and variation. In the Build It exhibit, as the components rotate, a set of handouts or pictures could be posted around the area to display potential ways to use the materials.

In addition to visual supports, expanding the use of multisensory communication strategies across exhibits may further strengthen engagement. While visual and tactile features were already present and effective in attracting attention, additional supports such as color-coding related components, incorporating tactile markers, or enhancing contrast may support understanding of how elements connect and function. Providing information through multiple sensory channels may support children with diverse sensory, cognitive, and communication needs while also reinforcing usability for a broader range of users.

Maintaining and continuing to support flexible, open-ended design is also recommended, as this was identified as a key strength across several exhibits. Many of the evaluated spaces already allowed for multiple approaches to interaction, supporting a range of play styles and levels of engagement. Preserving these qualities is important, as they enable children to participate according to their own interests, abilities, and pace without requiring a single method of interaction. In this way, flexibility not only reduces potential barriers but also expands opportunities for engagement, making it a critical consideration for future exhibit development.

Consideration of the physical positioning of exhibit components may further support independent participation. In several instances, children relied on caregivers to reach, reposition, or manipulate materials due to height, placement, or required force. Adjustments such as positioning components within a wider range of reach, providing alternative access points at varying heights, or incorporating elements that can be used from multiple positions may allow more children to engage independently while still supporting shared interaction. These changes do not require eliminating physically active play but rather ensuring that opportunities exist for engagement across a range of physical abilities.

In Build It specifically, there were instances observed where children and adults could not see or reach materials in the cubicles in the back. The positioning of the cubicles with limited lighting limited visual attention to that area and its materials. The tinker toys were arranged in a universally accessible manner with varying heights and open lids which attracted attention to those tools. Similar to this layout, having open toy storage stands to display other items present in the exhibit may attract and sustain engagement in that exhibit. Similar strategies may be applied across exhibits where materials are stored in centralized or less visible locations. The Gear Wall, for example, has all the components located in the storage box at the bottom on the floor. While this supports access for some users, children using seated mobility aids or with physical mobility restriction may not be able to reach down to grab these components. To address this, storage bins or tilted bins could be placed on the sides of the wall at different heights to create increased access for different users.

Spatial layout and circulation patterns may be considered in future exhibit modifications or planning. Observations indicated that certain areas became congested during periods of higher visitor activity, which limited ease of approach and interaction with exhibit components. Ensuring that materials can be accessed from multiple directions and that adequate space is maintained around high-use areas may support more consistent participation across varying conditions. Attention to how users move through and occupy space may also help reduce unintentional barriers that emerge during peak use.

Finally, while some strategies may require longer-term planning and alignment with funding and institutional priorities, consideration of larger-scale initiatives may further support overall access to and engagement within the museum environment. Expanding communication supports beyond standard visual signage presents one such opportunity. For example,

incorporating braille alongside essential and permanent exhibit text may support access for visitors with visual impairments, particularly when paired with clear visual contrast and simplified layouts. In addition, the integration of audio-based supports, such as auditory walkthroughs or exhibit narration available through the website or presented at the front desk, may provide alternative ways for visitors to receive and engage with information without reliance on printed text. These approaches have been identified as effective strategies for enhancing both accessibility and safety by supporting spatial awareness and independent navigation (Cieślik, 2024).

Extending these communication strategies further, existing museum resources provide a strong foundation for the development of more comprehensive supports. Building on current efforts, such as the social story available on the museum's website (Children's Museum in Oak Lawn, n.d.), the creation of sensory or audio-based stories that guide visitors through exhibits may further support preparation, understanding, and engagement. These resources can help familiarize visitors with the environment, reduce uncertainty, and provide flexible entry points into play experiences. Importantly, these supports do not replace open-ended play but instead offer optional layers of guidance that can be used as needed. Collectively, the inclusion of braille, audio supports, and sensory-based narratives reflects an approach that broadens access to information while maintaining flexibility in how visitors choose to engage with exhibits.

Overall, these recommendations build upon strengths already present within the museum environment while identifying opportunities to further support participation across a range of users. Incremental implementation of these strategies may help strengthen alignment with universal design principles while preserving the exploratory, collaborative, and open-ended nature of play within the museum. These recommendations reflect patterns identified across

Universal design principles, particularly in relation to intuitive use, perceptible information, and physical access.

9. Opportunities for Future Development

The findings of this review provide an initial exploration of how universal design principles may be reflected within selected exhibit environments at the Children's Museum in Oak Lawn. While this evaluation focused on four exhibits, the process also highlighted opportunities for future development related to the continued application of universal design principles, refinement of the evaluation tool, and expansion of evaluation efforts across the museum environment.

The continued refinement of the Universal Design Tool developed for this project is an important consideration for future work. As noted earlier, there is currently no standardized tool designed specifically for evaluating universal design within indoor children's museum play environments. Continued use and refinement of this tool should be informed by ongoing research on universal design as it relates to these constructs. This tool represents an initial effort informed by existing guidance and adapted to the context of play-based exhibits. The version used during this review was designed to support structured observation of environmental features aligned with the seven universal design principles. Through its application, several opportunities for refinement emerged, including clarifying observational prompts, reducing potential ambiguity in interpretation, and improving overall usability for future reviewers. Feedback from expert reviewers, along with practical experience using the tool in the museum setting, has already informed revisions. Continued refinement may further strengthen the tool's clarity, applicability, and usefulness for evaluating exhibit environments through a universal design lens.

Future development may also include expanding the use of the review tool to examine additional exhibits within the museum. Because this review focused on a limited number of spaces, applying the tool across a broader range of exhibits may provide a more comprehensive understanding of how universal design principles are reflected throughout the environment. This process may help identify broader patterns, strengths, and opportunities related to supporting participation and engagement across different exhibit types and visitor experiences.

In addition to evaluating existing exhibits, the review tool may serve as a resource for informing future exhibit planning and development. Incorporating universal design considerations during the early stages of exhibit design may support environments that are usable by a wider range of visitors from the outset, rather than requiring later modification. Applying a structured review process during concept development or prototype testing may also help identify potential barriers and opportunities for engagement prior to implementation.

At the same time, it is important to acknowledge that this tool was developed to evaluate individual exhibits rather than the museum environment as a whole. As a result, it does not fully capture broader environmental factors that influence participation across the full museum experience. The effectiveness of exhibit-level improvements may be limited if larger environmental considerations are not also addressed. Future efforts may include adapting this tool, developing a complementary assessment, or applying broader environmental universal design frameworks, such as those presented in the Universal Design New York guidelines, to better understand how the museum environment supports engagement across users.

Future work may also explore how evaluation processes informed by universal design can support ongoing reflection and collaboration among museum staff, designers, and community stakeholders. Periodic reviews may provide insight into how exhibits are used over

time and how visitor needs evolve, allowing for continued adaptation of the environment to support meaningful participation.

In addition to supporting participation, expanding usability and designing for a broader range of users may also have practical implications for the museum as an organization. Environments that are easier to navigate, understand, and engage with may attract a wider audience, including families who may not have previously considered the museum accessible to their needs. By supporting and intentionally targeting a broader range of visitors, these design approaches may contribute to increased attendance, repeat visitation, and overall community engagement, which are important considerations for the sustainability and growth of the museum.

Overall, this project demonstrates how applying universal design principles to the evaluation of museum exhibits can provide valuable insight into how play environments support access, usability, and engagement. Continued exploration of universal design evaluation and design practices may contribute to the development of children's museum environments that support meaningful play experiences, shared exploration, and participation across a wide range of users.

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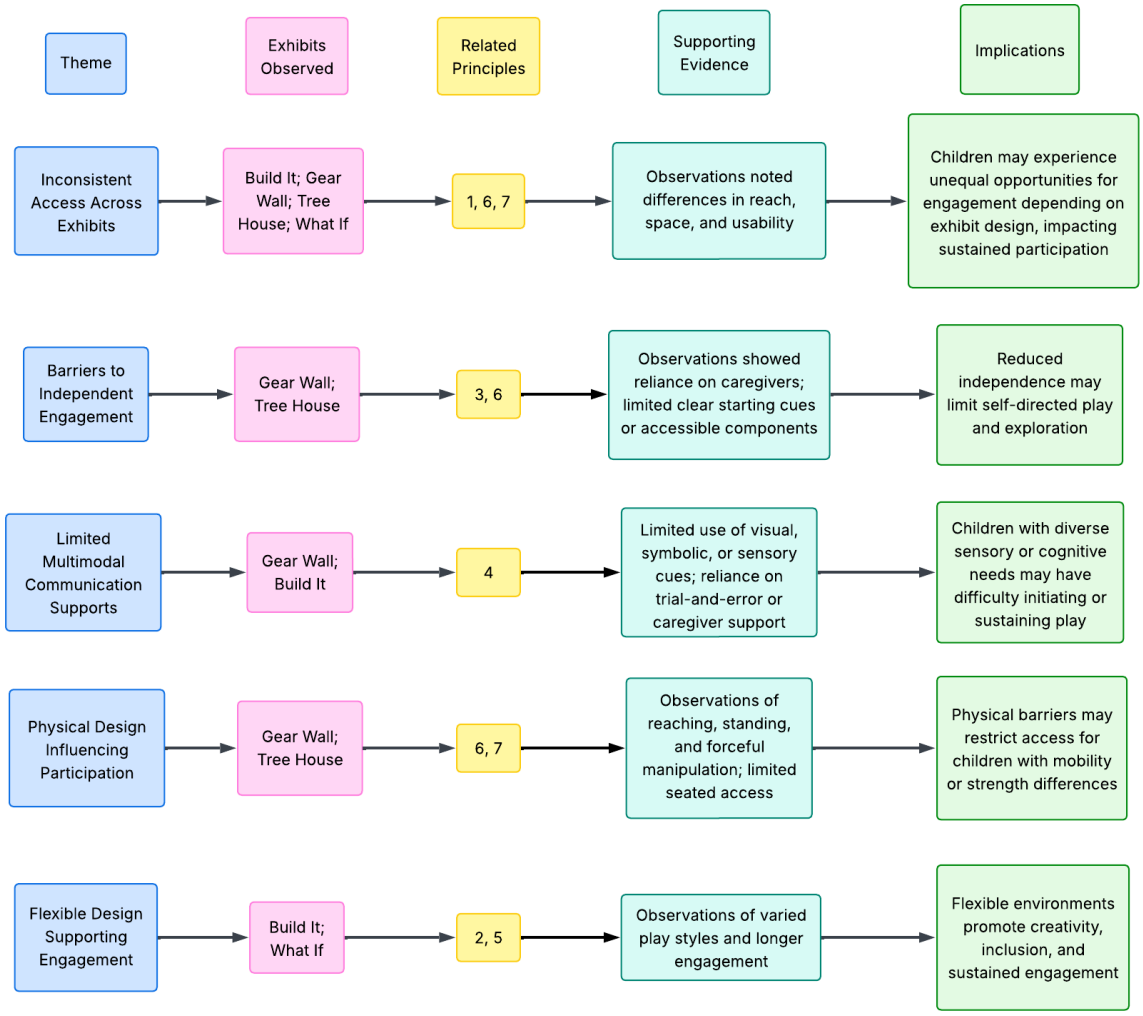
Appendix A

Thematic Matrix

Figure 1. Thematic Matrix from Results



Appendix B

Findings Tables

Table 1: UD Principle Ratings Across Observed Exhibits

UD Principle	Build It	Gear Wall	Tree House	What If?
Equitable Use	Moderate	Low–Moderate	Moderate	Moderate
Flexibility in Use	Moderate	Moderate	Strong	Moderate
Simple & Intuitive Use	Low	Low	Moderate	Low
Perceptible Information	Moderate	Moderate	Strong	Moderate
Tolerance for Error	Strong	Strong	Strong	Strong
Low Physical Effort	Moderate	Low	Moderate	Moderate
Size & Space for Approach	Moderate	Moderate	Moderate	Low

Table 2: Universal Design Principles Led Findings

Universal Design Principle	Observed Strengths	Observed Barriers	Implications
Equitable Use	Multiple children could often interact simultaneously; collaborative play encouraged	Some components positioned out of reach; occasional reliance on caregiver assistance	Some interactions may not support equal opportunities for engagement without assistance
Flexibility in Use	Many activities allowed open-ended exploration and multiple ways of interacting	Certain exhibit designs guided users toward more limited interaction pathways	Maintaining open-ended play supports diverse abilities and play styles
Simple & Intuitive Use	Children often able to explore materials independently through experimentation	Some exhibits lacked clear visual cues indicating how to begin interaction	Visual communication and intuitive design cues may improve independent engagement
Perceptible Information	Visual and tactile elements effectively attracted attention and supported initial interaction	Some information communicated primarily through written signage	Multi-sensory communication may support understanding and engagement across a wider range of users
Tolerance for Error	Trial-and-error exploration widely supported; components could be reset or reused	Few major barriers observed	Design supports experimentation and creative play

Low Physical Effort	Some components easy to manipulate	Certain materials required lifting, reaching, or caregiver assistance	Adjusting physical demands may support more independent and sustained participation
Size & Space for Approach and Use	Several exhibits allowed movement around components	Crowding occasionally limited access during busy periods	Spatial layout and circulation influence access and ease of interaction